

RODRIGO GOMES DE ARAÚJO

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EDUCATION

Master's in Computer Science

Faculdade de Ciências da Universidade do Porto (FCUP)

Sep 2025 to Present (Expected 2027)

Bachelor's in Computer Science and Engineering

Faculdade de Engenharia da Universidade do Porto (FEUP)

Sep 2022 to Jul 2025

RESEARCH

Cryptographically-Aware Session Types (CaST) | Master's thesis, *in progress*

2025 to Present

Supervisors: Pedro Ângelo and Bernardo Portela (FCUP) | *FreeST, Haskell, Type Systems, Applied Cryptography*

- Extending **context-free session types** (FreeST) so that a protocol's **cryptographic obligations** are tracked by the type system: settling a payment is **rejected at compile time** unless every authorization its context requires has been verified.
- Making the discipline **compositional over nested subprotocols** of arbitrary depth, preserving outer obligations across sequential composition, recursion, cancellation and crash-stop failure. Targets a **type-system extension** with a **soundness proof**, not a library; direct application to **smart contracts**.

PROJECTS

PoW Blockchain for Decentralized Auctions

Rust, Tokio, Dalek, libp2p, Tonic, Docker

- **Consensus and chain synchronization**, GossipSub block propagation, request-response recovery and a deterministic **longest-chain rule** for fork resolution.
- **Kademlia DHT overlay** (libp2p) for peer discovery, with **Blake2b** PoW, **Ed25519** signatures and **Merkle** transaction validation.
- **Decentralized auction application** preserving consistent world state across independent nodes.
- **Adversarial test harness** in Docker (miners, honest and malicious peers) validating the protocol against studied attack scenarios. **Graded 19.1/20** in Data Systems Security.

Elections data analysis and visualization

PostgreSQL, PostGIS, SQL, OCaml, Python, Data Warehousing

- **End-to-end electoral data platform**: ingestion, normalization, validation, warehousing and spatial analysis of official Portuguese election data.
- **Repeatable ETL pipeline** over heterogeneous official spreadsheets, handling varying election layouts and territorial and political-entity normalization.
- **SQL and PL/SQL analytical functions** for drill-down, vote swing, abstention, coalition handling and map-ready results, with an **OCaml** front end (Dream, Caqti, Bonsai). **Graded 18/20** in Advanced Topics in Databases.

ACHIEVEMENTS & AWARDS

2nd Place, IEEE RetroJam Game Development Competition | IEEE UP Student Branch | Rust

2025

- Developed a complete **2D game** within a **48-hour deadline** using Rust and Raylib.

Academic Merit Awards | Escola Secundária Carlos Amarante

2020 to 2022

World 4th Place, RoboCup Rescue Superteam | RoboCup Leipzig, Germany

2016

- Represented Portugal internationally in **autonomous robotics** competition, developing **navigation algorithms** and **autonomous systems** for **rescue** scenarios.

TECHNICAL SKILLS

Programming Languages: Rust, C/C++, Haskell, OCaml, Python, TypeScript, SQL

Systems & Networking: Tokio, libp2p, gRPC/Tonic, Dalek, POSIX, OpenMP

Data: PostgreSQL, PostGIS, SQLite, Caqti

Frameworks: Ratatui, Rodio, Raylib, Dream, Bonsai, FastAPI, React, Next.js, Node.js

Infrastructure & Tools: Git, Docker, Terraform, UNIX, Portainer, Cloudflare Tunnel, Tailscale

LANGUAGES

Portuguese: Native | **English:** B2 (Upper Intermediate) | **Spanish:** Proficient